FEDERAL SPECIFICATION

INSULATION BOARD, THERMAL, POLYURETHANE OR POLYISOCYANURATE FACED WITH MINERAL FIBERBOARD ON ONE SIDE OF THE FOAM AND GLASS MAT ON THE OTHER SIDE OF THE FOAM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

This specification forms a part of the latest issue of Federal Specification HH-I-1972/GEN.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for faced thermal insulation boards composed of rigid cellular polyurethane or polyisocyanurate faced with a mineral fiber board on one side of the foam and a glass mat facing on the other side of the foam. This specification does not cover cryogenic applications. For intended use see 6.1 of HH-I-1972/GEN.

1.2 Classification. The thermal insulation boards covered by this specification is composed of a rigid cellular polyurethane or polyisocyanurate foam faced with a mineral fiber board on one side of the foam and a glass mat facing on the other side of the foam and shall meet the physical requirements of 3.3, table I.

2. APPLICABLE DOCUMENTS

2.1 Latest issue of HH-I-1972/GEN and documents referenced therein.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C612 - Mineral Fiber Block and Board Thermal Insulation.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19102.)

FSC 5640

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3. REQUIREMENTS

3.1 Materials. The thermal insulation boards shall conform to the material requirements of the basic specification HH-I-1972/GEN and 3.1.1.

3.1.1 Facing materials. Facing materials incorporated in the design of the faced thermal insulation board, shall be as specified in 3.1.1.1 and 3.1.1.2.

3.1.1.1 Mineral fiber board. The mineral fiber board shall conform to the material and physical properties requirements specified in ASTM C612.

3.1.1.2 Glass fibrous mat. The glass fibrous mat shall be composed of glass fibers bound together with a suitable binder.

3.2 First article (see HH-I-1972/GEN).

3.3 Physical properties. The thermal insulation boards, when conditioned and tested in accordance with 4.5 through 4.5.7 of HH-I-1972/GEN shall meet the physical properties specified in table I.

TABLE I. Physical properties

Property	Value	Test Reqmt. Para.
Compressive strength (psi) minimum		
Rigid Cellular Polyurethane or polyisocyanurate portion of composite at 10 % deformation	16.0	4.5.3
Mineral fiber at 25 % deformation	12.0	4.5.3.1
Dimensional stability (L) at -40° F/ambient RH, (W) (percent change) maximum	2.0 2.0	4.5.4
Dimensional stability (L) at 158° F/97 % RH, (W) (percent change) maximum	4.0 4.0	4.5.4
Dimensional stability (L) at 200° F/ambient RH, (W) (percent change) maximum	4.0 4.0	4.5.4

4. QUALITY ASSURANCE PROVISIONS (see HH-I-1972/GEN)

5. PREPARATION FOR DELIVERY (see HH-I-1972/GEN)

6. NOTES

6.1 Intended use. The insulation material covered by this specification is intended for use as thermal insulation boards in roofs and other applications where the insulation will not be subjected to temperatures below -40° F or above 200° F. The service temperature will vary depending on the application. This specification does not cover cryogenic applications. Consult manufacturer for recommendations and physical properties of faced polyurethane or polyisocyanurate insulation boards in cryogenic conditions.

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6.2 Availability. Standard size - 3 feet by 4 feet.
Thickness - 1.5 to 3.6 inches

CUSTODIANS:

Army - ME
Navy - YD
Air Force - 99

Review activity:

Army - CE

CIVIL AGENCY COORDINATING ACTIVITIES:

COM - NBS
GSA - FSS, PCD

Preparing activity:

Navy - YD

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